Trainings

Setup

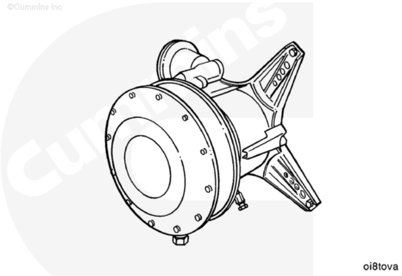
Engine Lifting Fixture, Part Number 3822512

Use engine lifting fixture, Part Number 3822512, to remove the engine from the chassis. Refer to Procedure 000-001 (/qs3/pubsys2/xml/en/procedures/40/40-000-001.html).

Install the engine to the test stand.

Align and connect the dynamometer. Refer to the manufacturer's instructions for aligning and testing the engine.

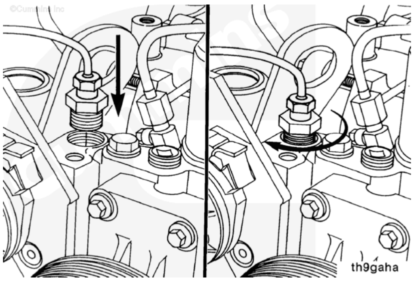
Note : Make sure the dynamometer capacity is sufficient to permit testing at 100 percent of the engine-rated horsepower. If the capacity is not enough, the testing procedure must be modified to the restrictions of the dynamometer.



Coolant Plumbing

Install the coolant temperature sensor.

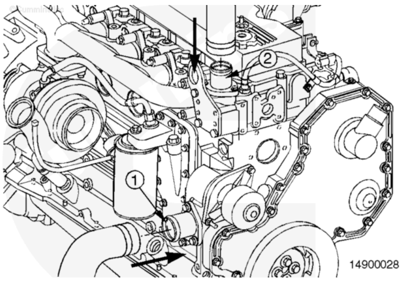
Measurements		
	celsius	fahrenheit
Minimum Gauge Capacity:	107	225



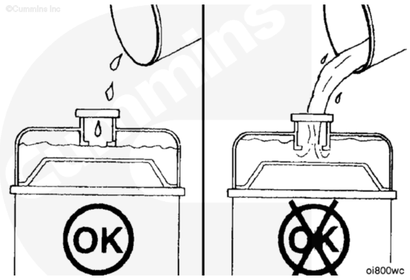
Connect the coolant supply to the water inlet connection (1).

Connect the coolant return to the water outlet connection (2).

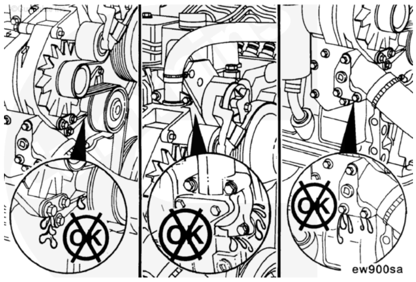
Install the drain plugs, close all the water drain cocks, and make sure all the clamps and fittings are tight.



Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.



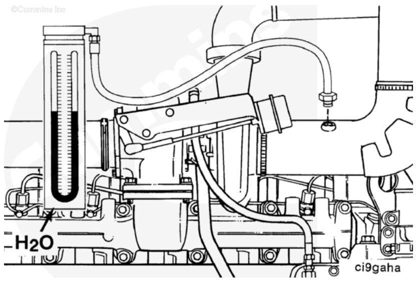
Inspect the engine for coolant leaks at connections, fittings, plates, and plugs. Repair as necessary.



Water Manometer, Part Number ST-1111-3

Air Inlet Restriction

Connect a water manometer, Part Number ST-1111-3, to the turbocharger air inlet pipe to test air restriction.



Note : The manometer connection must be installed at a 90-degree angle to the air flow in a straight section of pipe, one pipe diameter before the turbocharger.

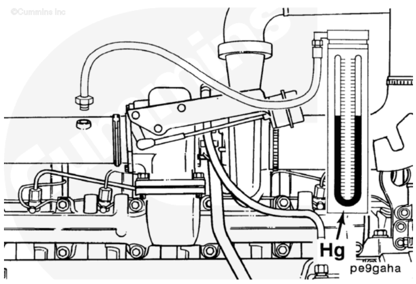
Note : A vacuum gauge, Part Number ST-434, can be used in place of the water manometer.

Measurements		
	mm-h2o	in-h2o
Minimum Gauge Capacity:	760	30

Pressure Gauge, Part Number ST-1273

Exhaust Restriction

Connect a mercury manometer to a straight section of the exhaust piping near the turbocharger outlet to check exhaust restriction.



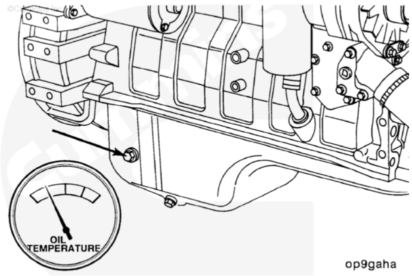
Note : A pressure gauge, Part Number ST-1273, can be used in place of the mercury manometer.

Note : For automotive applications a tapped hole is provided on the inlet side of the catalyst for checking

exhaust restrictions.

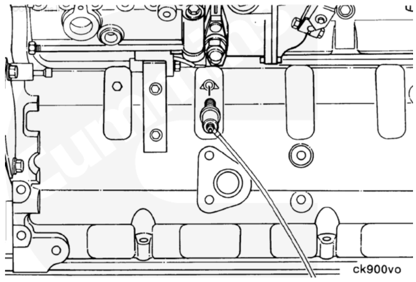
Measurements		
	mm-hg	in-hg
Minimum Gauge Capacity:	254	10

Attach the lubricating oil temperature sensor in the location shown.



Measurements		
	celsius	fahrenheit
Minimum Gauge Capacity:	150	302

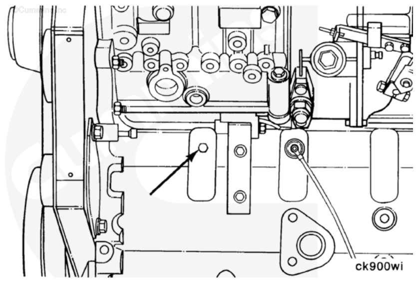
Attach the lubricating oil pressure sensor to the main oil rifle drilling in the cylinder block.



Measurements		
	kpa	psi
Minimum Gauge Capacity:	1034	150

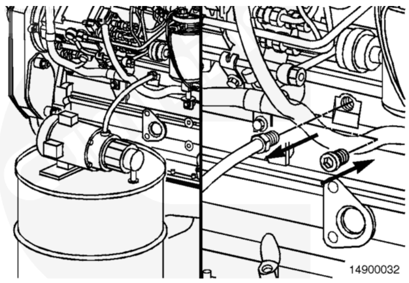
⚠ CAUTION ⚠

The lubricating oil system must be primed before operating the engine after it has been rebuilt to avoid internal damage.



To prime the system using external pressure, connect the supply to a tapped hole in the main lubricating oil rifle.

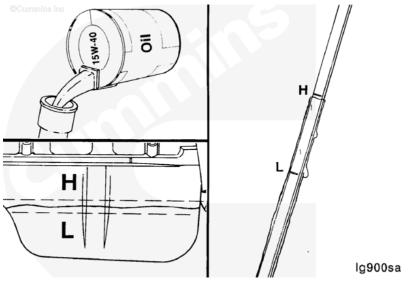
Use a pump capable of supplying 210 kPa [30 psi] of continuous pressure. Connect the pump to the port on the main lubricating oil rifle as shown.



Use clean lubricating engine oil to prime the system until the oil pressure registers on the gauge.

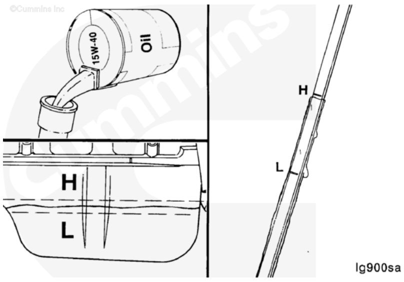
Remove the lubricating oil supply tube, and install the plug.

Make sure the lubricating oil has had time to drain to the lubricating oil pan, and fill the engine to the high mark as measured on the dipstick.



If an external pressure pump is **not** available, prime the lubricating system according to the following procedure.

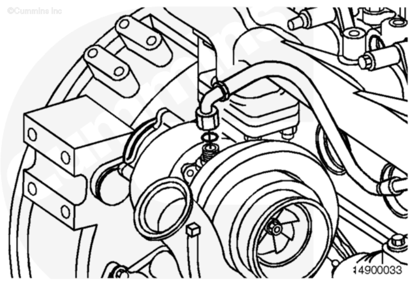
Fill the engine with lubricating oil to the high level mark on the dipstick.



Disconnect the turbocharger lubricating oil supply tube.

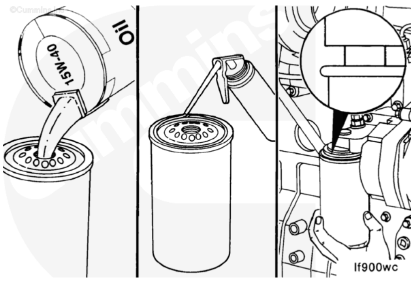
Pour 50 to 60 cc [2.0 to 3.0 fl oz] of clean lubricating engine oil into the turbocharger lubricating oil supply hole.

Connect the lubricating oil supply tube to the turbocharger.



⚠ CAUTION ⚠

Mechanical overtightening can distort the threads or damage the filter element seal.



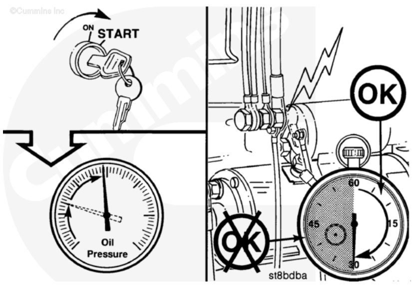
Fill the lubricating oil filters with clean lubricating engine oil.

Screw the filters onto the filter head fitting until the gasket contacts the filter head surface.

Tighten the filter as specified by the manufacturer.

⚠ CAUTION ⚠

Do not crank the starter motor for periods longer than 30 seconds. Excessive heat will damage the starter motor.



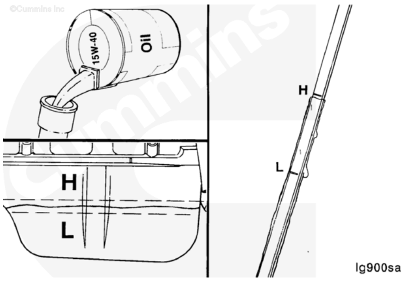
Crank the engine until the lubricating oil pressure gauge indicates system pressure.

Note : Allow 2 minutes between the 30-second cranking periods so the starter motor can cool.

Note : If pressure is not indicated, find and correct the problem before continuing.

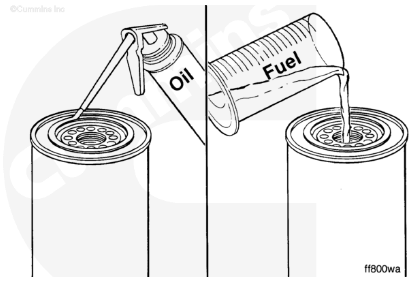
Allow the lubricating oil to drain into the lubricating oil pan, and measure the lubricating oil level with the dipstick.

Add lubricating oil, as necessary, to bring the level to the high level mark.



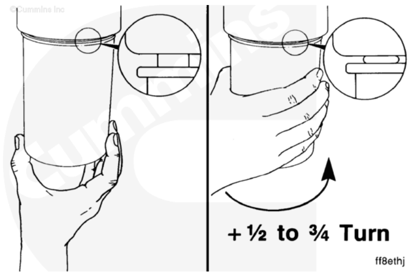
Lubricate the gasket on the fuel filter with clean lubricating engine oil.

Fill the fuel filter with clean fuel.



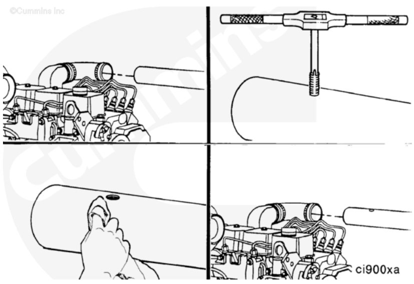
Screw the fuel filter onto the filter head until the gasket contacts the filter head surface.

Tighten the filter as specified by the manufacturer.



⚠ CAUTION ⚠

Do not attempt to install pipe thread fittings in plastic or rubber intake piping. Failure to do so will result in damage to threads.



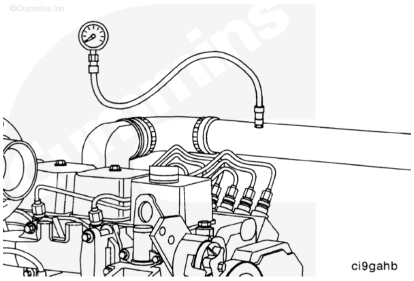
If the air crossover **not** have a pipe plug and tapped hole, perform the following procedure:

- Remove the crossover.
- Drill and tap a 1/8-inch pipe thread hole in the crossover.
- Clean all metal shavings from the air crossover.
- Install the crossover.

Pressure Gauge, Part Number ST-1273

To determine the amount of turbocharger boost, remove the pipe plug in the air crossover tube.

Install the intake manifold pressure sensor or pressure gauge, Part Number ST-1273.



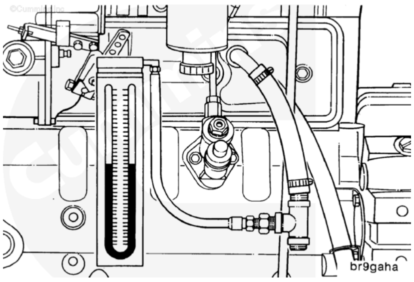
Measurements		
	mm-hg	in-hg
Minimum Gauge Capacity:	1905	75

Blowby Checking Tool, Part Number 3822476

Water Manometer, Part Number ST-1111-3

For accurate engine crankcase blowby measurement, insert a blowby checking tool in the crankcase breather vent.

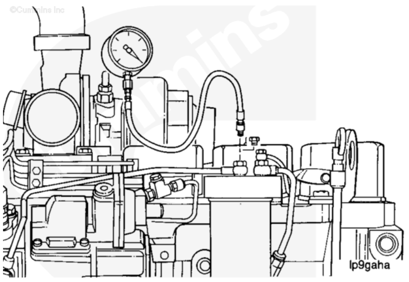
Connect a water manometer to the blowby tool. A pressure gauge can be used in place of the manometer.



Measurements		
	mm-h2o	in-h2o
Minimum Gauge Capacity:	1270	50

Vacuum Gauge, Part Number ST-434

To measure fuel filter restriction, connect vacuum gauge, Part Number ST-434, to the injection pump inlet line.



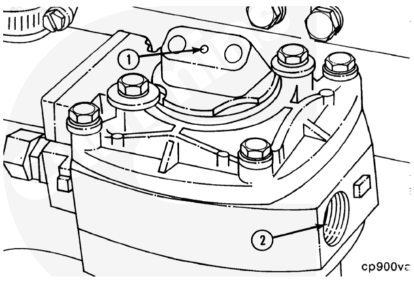
Measurements		
	mm-hg	in-hg
Minimum Gauge Capacity:	760	30

To be able to unload the compressor, connect a source of compressed air to the unloader (1). This air line **must** contain a valve between the source and the unloader.

Note : All air compressors manufactured by Cummins Inc. must be loaded during engine run-in. All air

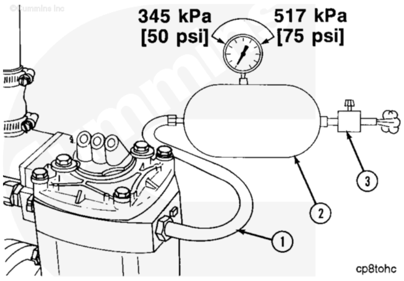
compressors must be unloaded during the engine performance check.

Note : The compressed air load in the accompanying illustration must be attached to the air compressor outlet (2).



To provide a load on the air compressor, connect an air tank (2) to the compressor outlet; use steel tubing or a high-temperature hose (1).

Install an air regulator (3) that can maintain tank air pressure of 345 kPa to 517 kPa [50 psi to 75 psi] at both the minimum and the maximum engine rpm.



Measurements		
	celsius	fahrenheit
Hose Temperature (Minimum):	260	500

Inspect the voltage rating on the starter motor before installing the electrical wiring.

Attach electrical wires to the starter motor and the batteries if used, negative (-) cable last.

Note : If another method of starting the engine is used, follow the manufacturer's instructions to make the necessary connections.